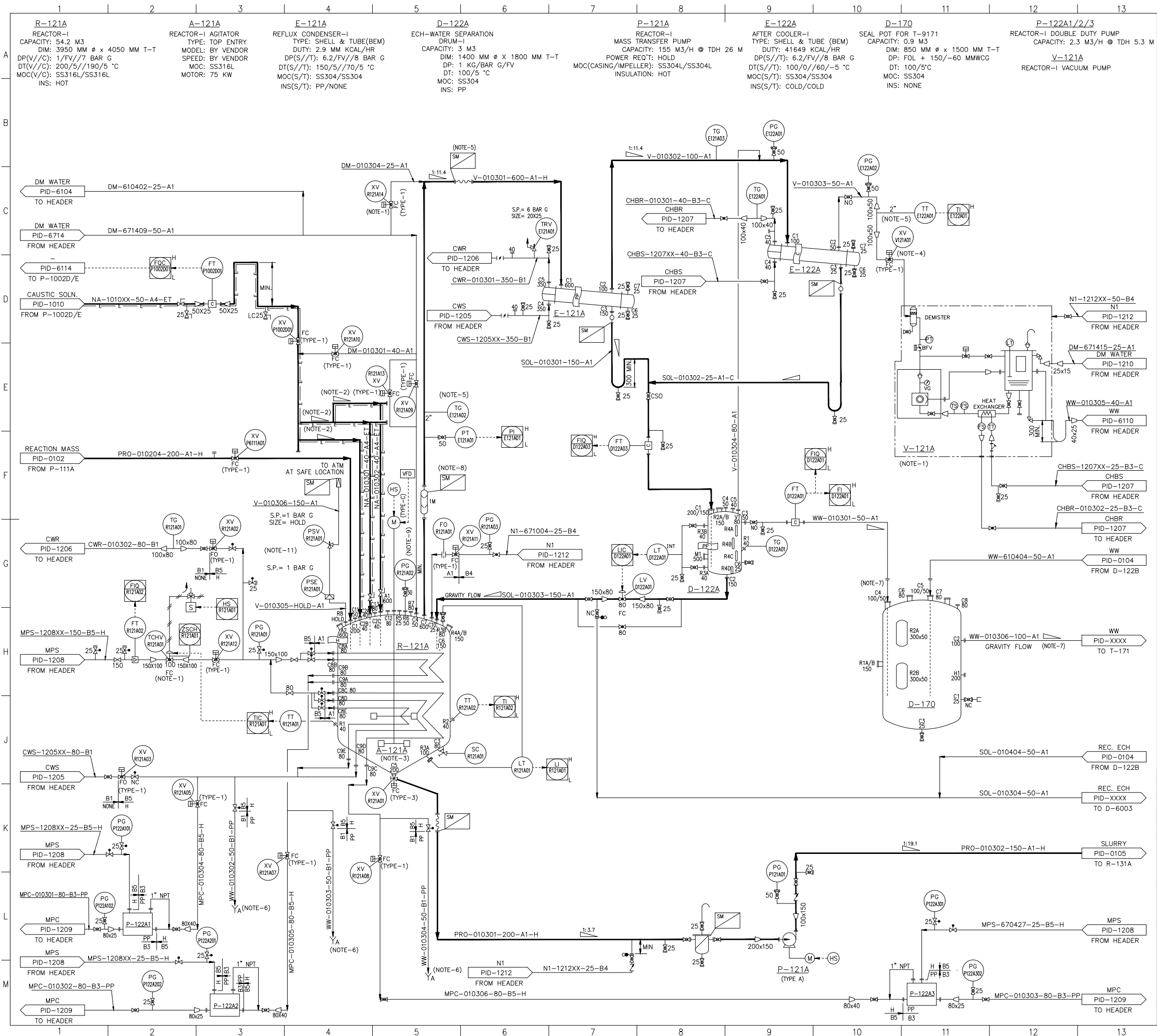
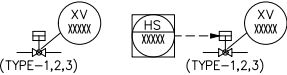




GENERAL NOTES:

1. ALL VAPOR LINES, SLURRY LINES AND GRAVITY LINES SHOULD HAVE MINIMUM BENDS AND NO LOOPS.
2. PROVIDE PROPER EARTHING AND BONDING FOR THE HYDROCARBON LINES.
3. USE LUBRICATED PLUG VALVES & PROVIDE FLANGE GUARDS FOR CAUSTIC SERVICE.
4. DRAIN TO BE PROVIDED AT THE LOWEST POINT AND VENT TO BE PROVIDED AT THE HIGHEST POINT.
5. REFER LEGEND SHEET FOR ABBREVIATIONS, SYMBOL, ON-O VALVE DETAILS, MOTOR/VFD AND STEAM TRAP DETAILS.
6. FOR LINES WITH SLOPE, SLOPE TO BE PROVIDED IN SUCH A WAY THAT THE LINE SHOULD BE DRAINED ON EITHER SIDE.
7. PROPOSED SPECIAL PIPING ITEM, TO BE FINALISED DURING DETAIL ENGINEERING.
8. TYPICAL XV SYMBOL ON PID TO BE READ AS BELOW.



9. FOR ALL INLINE INSTRUMENTS, UPSTREAM AND DOWNSTREAM DISTANCE TO BE CONSIDERED IN PIPING.
10. ORIFICE TO BE INSTALLED IN HORIZONTAL LINE ONLY.

NOTES:

1. FOR DETAILS, REFER VACUUM PUMP VENDOR P&ID TIPL-DES-02.
2. CAUSTIC SOLUTION INJECTION NOZZLES TO BE LOCATED AWAY FROM VAPOR VENT LINE TO E-121A.
3. AGITATOR MOTOR SHALL BE TRIP FROM MCC IN CASE OF MECHANICAL SEAL LEAKAGE.
4. DO NOT PROVIDE SLOPE IN THE LINE TOWARDS VACUUM PUMP.
5. THERMOWELL CONNECTION SIZE WILL BE 2" DUE TO HIGH VELOCITY IN LINE.
6. ALL DRAIN POINTS SHALL BE COLLECTED IN A SEPARATE HEADER AND TO BE SENT TO D-9303(PID#06105)
7. BREAK UP FLANGES TO BE PROVIDED IN DIP PIPE CONNECTED LINES.
8. SM-6186A(DEFLECTOR PLATE) TO BE MADE SUIT AT SITE BY EPIGRAL.
9. AGITATOR GEAR BOX SHALL BE PROVIDED WITH COOLING. WATER SUPPLY AND AGITATOR SEAL TO BE PROVIDED WITH DM WATER CONNECTION.
10. DM WATER FOR LINE FLUSHING.

HOLDS:

1. PSV & IN/OUT LINE SIZES.
2. PPPU PUMP DETAILS.
3. PUMP SUCTION-DISCHARGE NOZZLE SIZE.
4. VACUUM PUMP DETAILS.

A	13-APR-26	ISSUED FOR APPROVAL	DVS	NC	DP
No.	DATE	REVISION	DRAWN	CHD	REVD
					APPD



Rasayan Engineering Solutions

CLIENT:

EPIGRAL EPIGRAL Limited

PROJECT:

**BASIC ENGINEERING
FOR EPAM PROJECT**

TITLE:

**PIPING & INSTRUMENTATION DIAGRAM
REACTION SECTION (LER)**

SCALE : NTS DWG.NO.: EPAM01-PR-PID-0103 SHT: 1 OF 1 REV: A